BreedLog (by Kwantam)

Complete App Structure & Functional Specification (Full)

BreedLog is a private, offline-first livestock breeding and management mobile app for sheep farmers. Its core mission is to keep **bloodlines, breeding events, and family trees** accurate and usable on the farm — even with no internet. Data capture and improvement tracking are second. Evaluation support (manual scoring + optional AI valuation) is third.

Priority order (non■negotiable):

- **Breeding & bloodline genealogy**: sire/dam links, offspring, multi-generation pedigree trees, mating and lambing history.
- **Structured animal records**: identification, weights, performance/traits, reproduction, and health/treatments (LogiX-style depth).
- **Evaluation support**: 1–6 breed classing, suggestion logic, and optional AI valuation behind login.

Offline rule: The app must work fully in airplane mode for all core features. Only AI valuation requires internet.

1. Product Summary

BreedLog replaces notebooks and scattered spreadsheets with a single on-phone breeding register that preserves lineage integrity and decision history. A farmer can photograph an ewe, capture her data, link a chosen ram for breeding, record the lamb(s) at birth, and automatically build a living family tree.

Primary outcomes:

- Accurate sire–dam connections and multi-generation pedigrees as far back as the farmer’s data goes.
- Breeding event visibility: which ram covered which ewe, when, and what the lambing outcome was.
- A long-term herd improvement record: weights, traits, fertility, and health history per animal.
- Easy offline backups via CSV export, suitable for analysis and migration.

Target user: sheep farmers (Meatmaster focus), operating in low-connectivity environments.

2. Navigation & User Experience

The app is designed for outdoor use: high contrast, large tap targets, simple flows, and minimal steps. Bottom tabs keep the app predictable and “farm friendly.”

2.1 Bottom Tabs

- **Animals** — list, add/edit, profiles, photos, quick actions.
- **Breeding / Family Tree** — mating events, lambing events, pedigree trees, bloodline tracing.
- **Records & Performance** — weights, traits, reproduction reports, health treatments, history views.
- **AI Valuation** — login-gated; generate structured valuation text using OpenAI + save it locally.
- **Settings** — export/import, storage status, reset, app info.
- **Debug** (dev-only) — tests for logic + database integrity.

2.2 Visual Design System (Kwantam black/yellow)

Design token	Value
Background	#0B0B0B
Accent (Kwantam Yellow)	#FFC300
Cards / surfaces	#141414
Body text (off-white)	#F5F5F3
Headlines	#FFFFFF
Inputs (off-white)	#EDEDE9
Interaction style	Rounded cards, clean spacing, minimal icons, big butt

3. Core Workflows (Step-by-Step)

3.1 Add an Animal (Ewe/Ram/Lamb)

- User taps **Add Animal**.
- User captures or selects a **photo** (camera/gallery).
- User enters required ID fields (Tag ID; optional Electronic ID).
- User selects Sex (Ram / Ewe / Wether / Lamb).
- User enters birth date (YYYY-MM-DD) and optional birth weight.
- User assigns breed type (Euro / Afro / Middle) and other properties.
- User links Dam and Sire if known (search from existing animals).
- Save creates the animal record in local SQLite and stores photo URI/path.

3.2 Record a Breeding Event (Ewe × Ram)

- From an Ewe profile, user taps **Record Breeding**.
- Choose Ram (search + filter by active rams).
- Enter mating date and mating type (Natural / AI / Hand-mated).
- Optionally attach notes (camp, conditions, observation).
- Save creates a Breeding Event linked to ewe and ram.

3.3 Record Lambing & Create Offspring

- From the Breeding Event, user taps **Record Lambing**.
- Enter lambing date, number of lambs, survival status, notes.
- For each lamb: tap **Create Lamb Record**.
- App pre-fills Sire and Dam from the breeding event (non-editable by default; allow override only with confirmation).
- Assign Tag ID, sex, birth weight, notes, and photo per lamb.
- Save: lamb(s) are inserted and automatically linked to parents; generation number computed; family tree updates instantly.

3.4 View Family Tree / Pedigree

- From any animal profile, user taps **Family Tree**.
- Show parents, grandparents, and offspring branches.
- Allow expanding generations as far back as data exists (lazy-load to keep it fast).
- Provide quick links: open profile for any ancestor/offspring.
- Provide an optional **Inbreeding warning** placeholder: flag when selected sire appears in dam's ancestry (simple detection; full coefficient can be later).

3.5 Export Data (Backups)

- User opens Settings → **Export CSV**.
- App exports multiple CSV files (one per table) and offers share/save.
- Exports must include relationships: animal IDs + sire/dam IDs, breeding event IDs, offspring links.

4. Animal Recording Fields (Structured & Attractive)

BreedLog must capture deep livestock data similar in spirit to established breeding systems (e.g., LogiX-style records), but without relying on internet or external registries. The farmer must be able to build the dataset gradually.

4.1 Animal Information

- **Identification**: Tag ID (required), Electronic ID (optional), Name (optional), Breed (default Meatmaster).
- **Birth**: Birth date, Birth weight, Birth type (single/twin/triplet optional), Birth notes.
- **Sex**: Ram / Ewe / Wether (and Lamb as a status or derived from age).
- **Ownership**: Breeder, Owner, Farm name, Location note (non-GPS).
- **Status**: Active, Sold, Dead, Culled, Lost (with date + reason).
- **Photos**: profile photo + optional gallery images (future).
- **Notes**: free-text notes for anything not covered.

4.2 Pedigree Information (Multi-Generation)

- Sire link (points to another Animal record).
- Dam link (points to another Animal record).
- Offspring list auto-populated based on sire/dam links and offspring table.
- Pedigree viewer supports multi-generation expansion.

4.3 Performance Data & Traits

- Weights: birth, weaning, 6 months, 12 months, plus ad-hoc weights with dates.
- Growth/performance notes and simple metrics (e.g., ADG placeholder).
- Traits/properties: constitution note, adaptability note, feet/legs note, pigmentation note (extensible).
- EBV/GEBV: support manual entry fields as placeholders (no calculations required initially).

4.4 Health & Reproduction

- Health treatments: date, treatment, medication, dosage, withdrawal note (optional), vet/handler, notes.
- Reproduction: mating lists, lambing outcomes, fertility observations, assisted birth notes, mothering notes.
- Fertility reports (simple summaries): number of matings, conceptions, lambing %, lambs weaned (future).

5. Evaluation & Improvement Tools (Tertiary)

Evaluation supports improvement decisions but must not replace breeding record accuracy. All evaluations are stored locally and can be linked to animals over time.

5.1 Manual Breed Classing (1–6 scale)

- Head score (1–6)
- Front Quarter score (1–6)
- Middle Section score (1–6)
- Rear Quarter score (1–6)
- Overall Type: Euro / Afro / Middle
- Comment: official selection language (stored as text).
- Multiple evaluations per animal allowed; latest is shown on profile.

5.2 Suggestion Box (Ewe lamb ≥ 6 months)

Exact rules (must match):

- If ageMonths < 6 → not eligible.
- avg = (head + front + middle + rear) / 4
- rec = type ?? Middle
- If rear ≤ 3 OR middle ≤ 3 → rec = Euro
- If head ≤ 3 OR front ≤ 3 → rec = Afro
- If avg between 3.5 and 4.5 inclusive → rec = Middle

Rationales:

- Euro: Select a Euro-type ram to add hindquarter and loin expression, depth and early growth.
- Afro: Select an Afro-type ram to strengthen adaptation, constitution, hooves and foraging efficiency.
- Middle: Use a well-balanced middle-of-the-road ram to consolidate type without over-correcting.

6. AI Valuation (Login-Gated, Optional)

AI valuation is an optional productivity feature used to generate professional narrative evaluations. It must be gated behind login because it uses internet and paid API calls. The core breeding register remains fully offline and usable without login.

6.1 Login & Security

- AI tab shows Login screen if not authenticated.
- Login fields: Email/User ID + Password.
- Use secure storage for session token/credentials (platform secure store).
- Include logout.
- If offline: AI screen explains internet required, but offline records remain available.

6.2 AI Valuation Workflow

- User selects an animal (search by Tag ID).
- App composes a structured prompt using: animal details + lineage links + latest evaluation scores + notes.
- User taps **Generate Valuation**.
- AI returns a structured report with sections: Head, Front Quarter, Middle Section, Rear Quarter, Overall Type, Improvement/Breeding recommendation.
- User can: Copy to clipboard, Save to Animal (stored locally as AI Valuation record).
- Add rate limiting/cooldown to prevent spam.

6.3 OpenAI Integration Requirements

- Use Replit integrations/secret management (no hardcoded API keys).
- Gracefully handle errors: timeouts, rate limits, invalid key, no internet.
- Never block core offline features if AI fails.

7. Offline Rules, Export/Import, Data Safety

7.1 Offline-First Rules

- All core CRUD must work with no internet.
- Database is local SQLite; no cloud sync required.
- Photos stored on device and referenced in DB by URI/path.
- AI features are optional and do not affect core records.

7.2 Export CSV (Mandatory)

Export must create one file per table (minimum):

- animals.csv
- breeding_events.csv
- offspring.csv
- performance_records.csv
- health_records.csv
- evaluations.csv
- ai_valuations.csv

Export requirements:

- Include primary keys and foreign keys so relationships can be reconstructed.
- Offer Share/Save after export (device share sheet).
- Show a clear success message and file locations.

7.3 Import CSV (Strongly Preferred)

- Allow selecting CSV files and importing them back into SQLite.
- Validate columns; show friendly errors.
- Avoid duplicates by Tag ID when possible (prompt user on conflicts).

8. Data Integrity, Validation & Edge Cases

- Tag ID required; prevent empty tag saves.
- If Tag ID unique: warn on duplicates; allow override only with confirmation.
- Parent links: prevent an animal being its own ancestor (basic cycle check).
- Breeding event requires ewe + ram + mating date.
- Lamb creation from breeding event must auto-assign sire/dam and warn if user overrides.
- Deleting animals must warn if it is a parent of other animals; offer safe options: mark Sold/Dead vs delete.
- Foreign keys enabled; cascade rules used intentionally (scores/ai valuations to cascade; lineage deletions must be cautious).

9. Debug Screen (Dev Only) — Required Tests

9.1 Logic Tests

- Weak middle/rear → recommends Euro.
- Weak head/front → recommends Afro.
- All 4s → recommends Middle.
- Age 5 months → not eligible.

9.2 Database Integration Tests

- Insert animal → ID returned.
- Fetch animal → matches.
- Insert breeding event → links correctly.
- Create lamb from event → sire/dam links correct.
- Insert evaluation → fetch latest evaluation.
- Delete animal (safe test record) → related eval/ai valuation removed as expected.
- Confirm family tree function returns expected ancestors/descendants for the test graph.

10. Future Enhancements (Optional Roadmap)

- Inbreeding coefficient calculations and warnings (full pedigree-based).
- Batch actions: bulk weigh, bulk treatments, bulk lambing capture.
- Breeding season planning tools (ram groups, camps, mating lists).
- Reports: fertility % per ewe, lamb survival, growth curves, ram performance dashboard.
- Photo gallery per animal and attachment of documents (vaccination cards).
- Encrypted local backups and/or optional cloud sync (user-controlled).

Summary: BreedLog is a breeding-first, offline livestock register that preserves bloodlines and decision history. Data recording is deep and structured. Evaluation is supportive, with AI as an optional, login-gated tool.